

Fault diagnosis in photovoltaic systems using the Fourier transform

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SUMMARY

This paper presents a study of fault diagnosis in photovoltaic systems using the Fourier Transform, which involves analyzing the frequency patterns of the electrical waveforms generated by solar panels. This technique makes it possible to identify problems such as hot spots, partial shading and other defects that affect the performance and efficiency of the systems. The application of the Fourier Transform provides a powerful tool for the early detection of faults, contributing to preventive maintenance and increasing the useful life of photovoltaic systems.

I. INTRODUCTION

Panels for photovoltaic power generation are increasingly being used for distributed micro-generation purposes and also to supply electricity in locations far from high-voltage power lines[1], [2].

Its useful life has reached the point where its high installation cost is worthwhile. One difficulty with its maintenance when installed in hard-to-reach places is the replacement of defective elements [3], [4].

Faults in the manufacture of solar panels can result from inadequate processes, poor quality control or defective materials, such as variations in the thickness of the photovoltaic layer, the presence of impurities and poorly welded connections. These concrete problems have the potential to significantly compromise the panel's efficiency and reduce its lifespan. In a scenario of growing demand for solar energy[5]

II. LITERATURE REVIEW

A. PV system failures

PV generation is recognized as one of the least maintenance-intensive forms of generation. However, the panels are generally exposed to the external environment, which entails a considerable probability of electrical faults occurring and their safety, reliability and efficiency are susceptible to these faults [6].

In addition, the quest to reduce the price of panels has led manufacturers to worry less and less about quality and most failures occur due to problems caused by the use of low-quality materials and manufacturing processes [7].

This can significantly reduce (or even cancel out) energy generation and also increase the risk of serious problems such as fires [8]. Figure 1 describes the classification of various defects that define faults in PV systems on both the direct current (DC) and alternating current sides.(CA) [8].

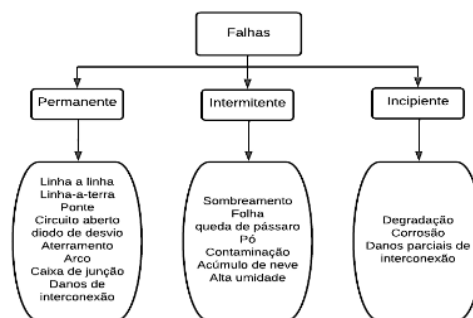


Fig.1. Common PV system faults

Fig. 1 shows the main faults that occur in PV systems. Generally, the DC side of the panels is always more affected than the AC side. One of the most catastrophic failures on this side is hot spot failure, which can occur due to partial shading of the panel [9].

There are also wear faults, which are usually visible as lighter, roughly circular marks. These marks are also sharper in infrared spectrum images, as in Fig.2, because the points of inefficient or non-existent conversion of sunlight must heat up more. In other words, the absence of the photoelectric effect, which directs the energy of the light photons, means that more energy is converted into heat[10].

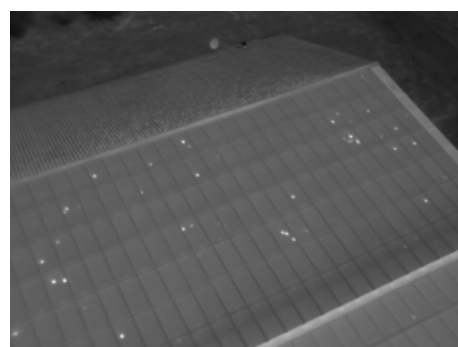


Fig. 2. Visible faults in a photovoltaic panel using infrared spectrum photography

III. FAULT DIAGNOSIS IN PHOTOVOLTAIC SYSTEMS USING THE FOURIER TRANSFORM

The Fast Fourier Transform (FFT) power spectral analysis technique is often used to evaluate the operating condition of power electronic systems[11]. Examples of applications of this method include: detecting broken bars in three-phase induction motors in industrial and traction systems, faults in IGBTs in frequency inverters, among others. In order to apply the method, it is necessary to understand the sampling and the Fourier transform are described briefly and objectively below.

B. The Fourier transform and the sampling theorem

From a signal represented in the time domain, the Fourier transform makes it possible to calculate the spectrum of the signal, which is the representation of that signal in the frequency domain. Alternatively, starting from a representation in the frequency domain, it is possible to obtain a representation of the signal in the time domain using the inverse Fourier transform. The discrete-time Fourier transform of a sequence $X(n)$ can be defined by the following equation:

$$(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} X(n) e^{-j\omega n} \quad (1)$$

It is worth remembering that $X(e^{j\omega})$, if convergent, results in a function of ω which is continuous or generally continuous, and which is also periodic in ω with a period of 2π . The periodicity of the function can be seen from the fact that $e^{j(\omega+2\pi)} = e^{j\omega}$. The discrete-time Fourier transform has an inverse operation, which transforms $X(e^{j\omega})$ back into $X(n)$, as shown in Eq. (2)

$$X(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad (2)$$

IV. RESULTS

Spectral analysis refers to the study of the properties of a signal or system in the frequency domain. Specifically, spectral analysis involves breaking down a signal into its frequency components in order to better understand its characteristics. This is done using techniques such as the Fourier Transform, which converts a signal from the time domain to the frequency domain, showing which frequencies are present and with what intensity.

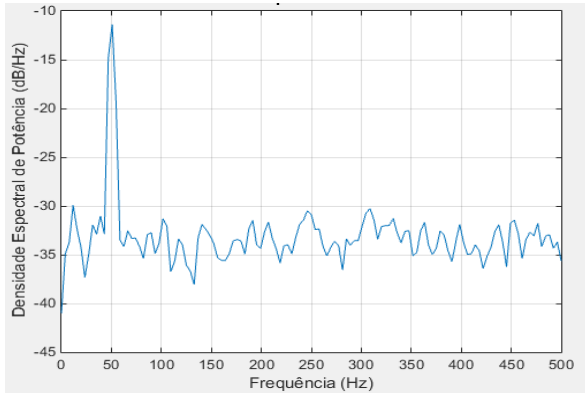


Fig. 3. Power spectral analysis

Power versus Voltage Curve and the Fourier Transform Although the I-V characteristic curves of PV arrays are neither periodic nor a direct function of time, it is possible, considering the effects discussed above, to treat them as a periodic wave. In order to apply spectral analysis using traditional Fourier transform (FFT) algorithms, the I-V characteristic curve has undergone a pre-processing stage Due to noise and errors present in the curve sampling process, it is not possible to guarantee that for each voltage value there is a single corresponding current value, so it is necessary to obtain a set of equally spaced points.

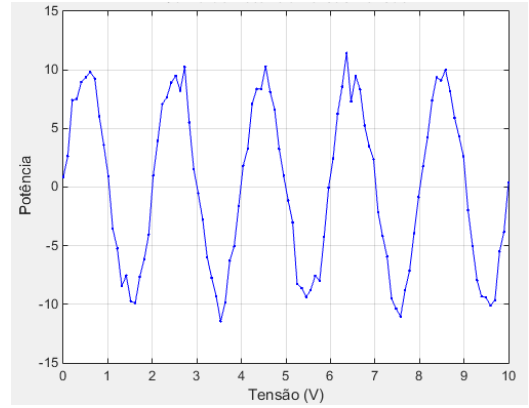


Fig. 4. Power x voltage

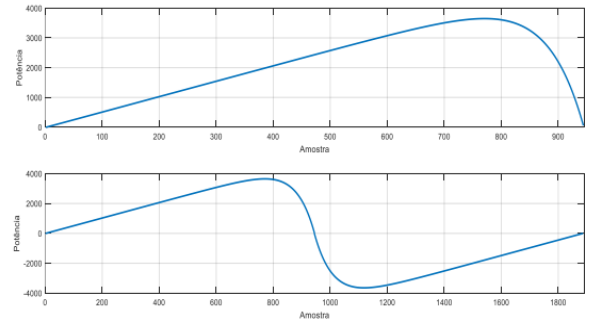


Fig. 5. Original P-V curve and curve used to calculate the Fourier transform

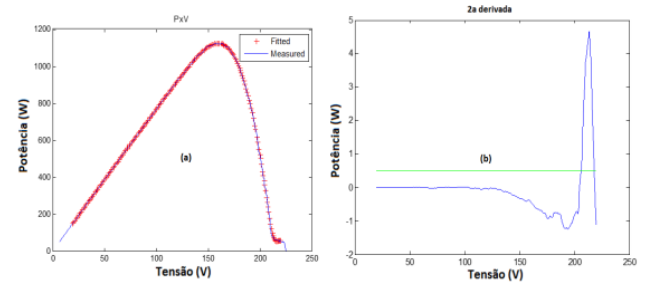


Fig. 6 (a) Adjustment of a P-V curve operating with partial shading; (b) second derivative of the P-V curve under conditions with partial shading.

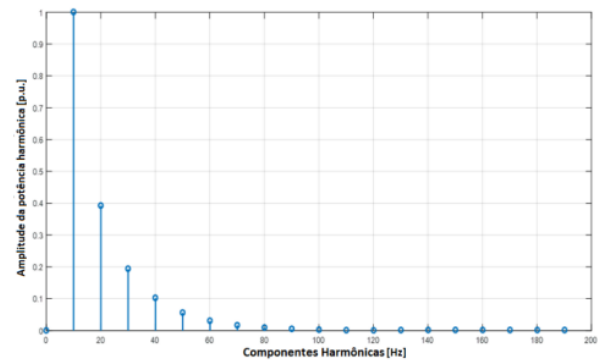


Fig. 7. Harmonic power spectrum (fault diagnosis) for measurement 1 in normal operation.

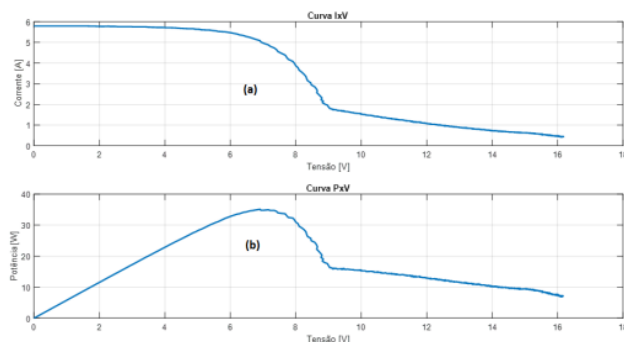


Fig. 8 (a) I x V curve (b) P x V curve: both under partially shaded operating conditions.

CONCLUSION

Photovoltaic systems have certain faults which can affect their operation and the useful life of the components. There are therefore a number of methods used to diagnose these faults, as in our case, these faults were diagnosed using the Fourier transform. Finally, both methods proved to be effective in detecting faults in photovoltaic systems, but further tests are needed to deepen the studies/analyses and thus correct errors, optimize the algorithms and the accuracy of each method to better diagnose possible faults in photovoltaic systems [11].

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